

Application Note: Timed Fire Events

CAUTION

When using FreeETarget:

- Wear eye protection
- Maintain a safe distance
- Do not shoot into any areas except the open target areas.

The freeETarget project or contributors will not be responsible for any injuries when using this target or its components.

Target shooting can be dangerous so apply caution in everything you do.

SUMMARY

In addition to the conventional course of fire, FreeETarget supports timed events

- Rapid Fire Pistol,
- Rapid Fire Sport Pistol
- Tabata Training

Required

The following are required to use the timed events with FreeETarget

- PC Client, Version 4.12.2 or higher
- Target Firmware 6.4.2 or higher

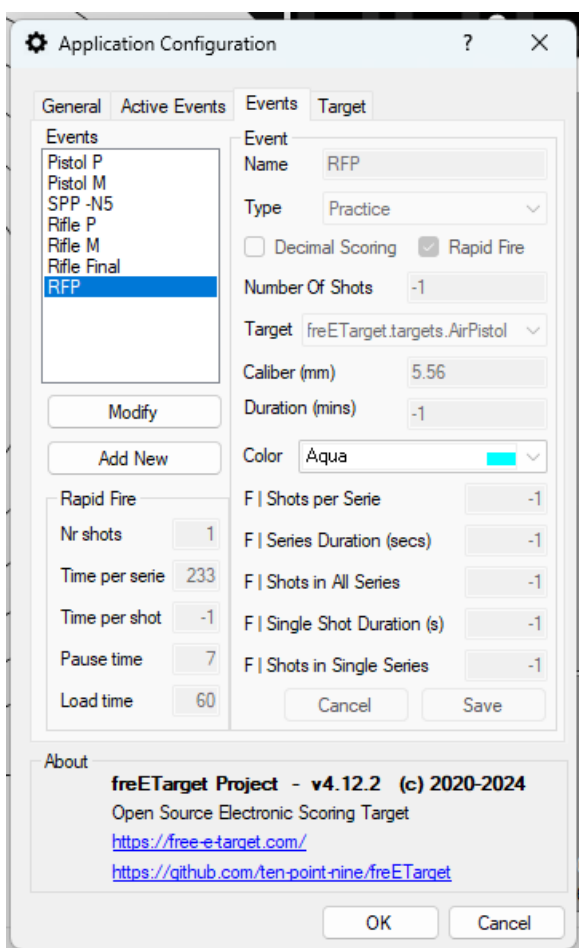
Optional

- Timed Fire Traffic Lights <https://free-e-target.com/service-documents/#TimedFireTrafficLights>

Rapid Fire Pistol

Rapid fire pistol is the event where the athlete has a short period of time to fire a number of shots into the target. The event is controlled by the Events Tab in the settings.

Figure 1 shows the settings for the Rapid Fire Pistol



Name: RFP

Target: Choose the target used for this relay

Check Rapid Fire

Nr shots. The number of times the event is repeated. Typically set to 1

(Note: Each series is typically 5 shots long)

Time per series. How long the event runs in seconds.

Pause time. How long the athlete is warned before the event time is active

Load time. How long the athlete is given to prepare for the event

Figure 1: Rapid Fire Pistol Settings

Typical Values

Event	Nr shots	Time per series	Pause time	Load time	Shot Override
NRA Timed Fire	1	60	10	60	-N10
NRA Rapid Fire	1	10	10	60	-N10
ISSF Timed Fire	1	233	10	60	

Sport Rapid Fire Pistol

Sport Rapid Fire Pistol differs from Rapid fire pistol in that the athlete is presented with a number of targets all spaced apart in time and they have a short period of time to put one shot into the target.

Figure 2 shows the settings for the Sport Rapid Fire Pistol

The screenshot shows the 'Application Configuration' window for 'freETarget Project - v4.12.2 (c) 2020-2024'. The 'Events' tab is active, and 'SPP -N5' is selected in the 'Events' list. The 'Event' details are as follows:

- Name: SPP -N5
- Type: Final
- Decimal Scoring: ☒
- Rapid Fire: ☒
- Number Of Shots: 5
- Target: freETarget.targets.AirPistol
- Caliber (mm): 5.56
- Duration (mins): -1
- Color: Red
- F | Shots per Serie: 5
- F | Series Duration (secs): 250
- F | Shots in All Series: 10
- F | Single Shot Duration (s): 50
- F | Shots in Single Series: 2

Below the 'Event' details, there are 'Cancel' and 'Save' buttons. At the bottom of the window, there is an 'About' section with the project name, version, and copyright information, along with links to the project's website and GitHub repository.

Name: RFP

Target: Choose the target used for this relay

Check Rapid Fire

Nr shots. The number of times the event is repeated. Typically set to 1

(Note: Each series is typically 5 shots long)

Time per series. How long the event runs in seconds.

Pause time. How long the athlete is warned before the event time is active

Load time. How long the athlete is given to prepare for the event

In this example, the athlete has 50 seconds to lay down 5 shots. This is made up as

$50 = 5 \times (7 \text{ second pause} + 3 \text{ seconds live})$

The 3 second live time is calculated based on the 5, 50, and 7 settings

Figure 2: Sport Rapid Fire Pistol Settings

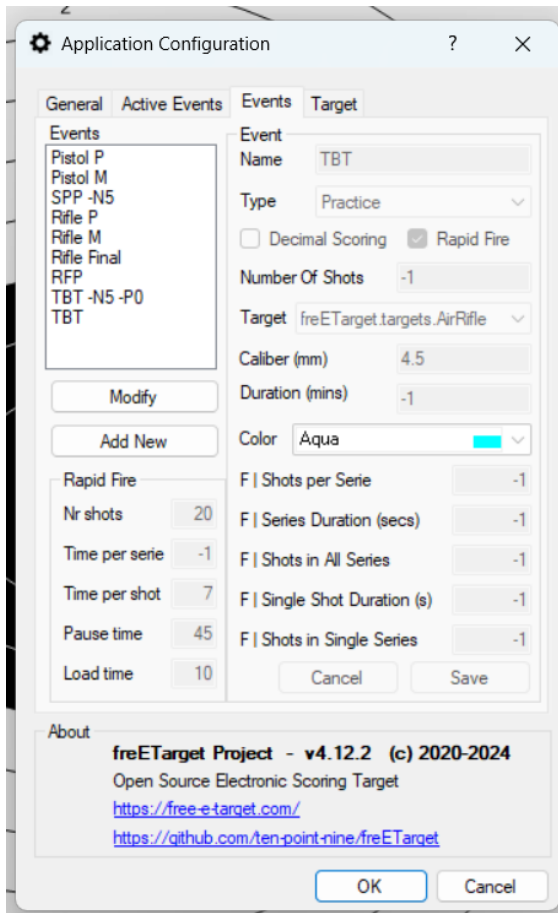
Typical Values

Event	Nr shots	Time per series	Pause time	Load time	Shot Override
Timed Fire	1	50	7	60	-N5

Tabata

Tabata is a training aid to control the amount of time an athlete has on the target to make a shot.

Figure 3 shows the settings for the Tabata



Name: TBT

Target: Choose the target used for this relay

Check Rapid Fire

Nr shots. The number of times the event is repeated. Typically set to 20

(Note: Each series is 1 shot long)

Time per shot. How long the target is active for. Typically less than 7 seconds.

Pause time. How long between shots

Load time. How long the athlete is given to prepare for the event

In this example, the athlete will be presented with 20 shots that are active for 7 seconds, and then a 45 second break to reload and prepare.

Figure 3: Tabata Settings

Overrides

You can tailor each event by adding overrides to its name.

Overrides are applied to the event name, for example, SPP -N6 allows six shots instead of the standard five.

Overrides may be applied to ANY event to create special purpose training events.

The overrides are presented below.

Override	Description	Purpose	Example
-N	Number of shots in the event	Tailor the number of shots to a specific event	-N5 Five shots will be fired in one event
-P	Override {"PAPER_TIME":0}	Temporarily disable the witness paper drive during the event	-P0 Do not advance the paper when a shot is fired
-E	Override {"PAPER_ECO":0}	Expand the PAPER_ECO for the event	-E30 Temporarily set the PAPER_ECO to 30mm